Small-UAS Navigation using 3D Imager and Infrared Camera in Structured Environments

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***Abstract*— This paper discusses a positioning and navigation algorithm to support autonomous flight of a small-size unmanned aircraft system in a structured GNSS-denied environment during a maintenance mission. The proposed method is based on the integration of depth imagery and outputs from an inertial measurement unit (IMU). In addition, 2D images of a low resolution long-wave IR (LWIR) sensor are used to aid the navigation process and support the maintenance mission by creating a thermal map of the indoor environment. A platform has been built with the appropriate sensors and data has been collected. The paper describes the methodology, the test platform and some preliminary results from mostly the preprocessing steps.**

Keywords—sensor integration; GNSS-denied environment; inertial navigation; electro-optical sensors; depth imagery; iterative closest point; simultaneous localization and mapping.

I. INTRODUCTION

Unmanned Aircraft Systems (UAS) are proving to be increasingly favorable in military and commercial applications. The range of applications include surveillance, aerial photography, environmental observations, search and rescue, mapping, forestry/agricultural survey, law enforcement among many. The small size unmanned multi-copters are highly capable and cost effective for low altitude operations and have extended access to dangerous and hazardous environments which were previously unavailable. Irrespective of the applications, a position and navigation solution is necessary to fly the UAS completely autonomous or even to manually control it easily. The Global Navigation Satellite System (GNSS) has become one of the most dependable solution for position and navigation outdoors but does not perform well in the indoor environment as the signal is obstructed by the roof and the walls. Hence, there is a need for non-GNSS position and navigation solution methods for indoor environments. Simultaneous Localization and Mapping (SLAM) and feature-based integrated navigation are two methods that can be used for this purpose using various types of sensors like ranging sensors, cameras, and inertial measurement units (IMU). This paper will focus on integrating depth imagery and low-resolution LWIR imagery with an IMU to obtain and estimate of both the position and the map of the environment. In this discussion, the region of operation is restricted to structured environments and would be extended to unstructured environments in the future.

This paper makes use of depth imagery where each image pixel is associated with an intensity value and a 3D point cloud value. Each pixel can, thus, be represented by a vector (x, y, z, η). One category of sensors that is capable of acquiring depth imagery is based on the principle of Time of Flight (TOF), i.e. by measuring the time-of-flight for light emitted from the camera to reach a target and be reflected back. Examples of TOF sensors are active 3D camera systems that use the transmission and reception of pulses such as the 3D flash LIDAR products by Ascending Technologies, or sensors based on the transmission or reception of amplitude modulated Infrared (IR) such as the Swissranger and PMD sensors. The use of an active sensor to create a 3D images overcomes the inability of a passive stereoscopic 3D image sensors to estimate the distance to an object in a scene when there is insufficient contrast between the object and the background. More recent sensors that produce depth images are the Kinect and the Occipital. The principle of operation of both these sensors is based on structured light [1].

Two-dimensional (2D) laser scanners have been and are still extensively used for navigation of robots in a structured indoor environment. For example, [2] uses a set of extracted lines and points from a 2D sensor to compute the translation and rotation of a robot platform [3]. The idea of using planar surfaces extracted from gimbaled 2D laser scanners for 2D localization is described in detail in [4]. In [5], the author extends the use of 3D imagery from a gimbaled 2D laser scanner to simultaneous localization and mapping (SLAM). Use of 3D features in Flash Ladar imagery was introduced in [6] and [7].



Figure 1. Target environment for maintenance operations.

Within the context of this paper, the 3D imagery is used in combination with LWIR imagery to enable navigation of a small-size UAS in maintenance tunnels where a typical GNSS-

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based position capability is unavailable. Challenging environments are divided into two types of environments: (i) the "structured" environment characterized by well-defined parameters such as predictable ceiling heights, room/corridor shapes/sizes, and standard building materials, (ii) and the "unstructured" environment characterized by irregular dimensions and rough surfaces [8]. The maintenance tunnel environment targeted in this paper is an example of a structured environment and shown in Figure 1.

The proposed method performs navigation using features present in 3D imagery, the point cloud data itself and, if present, features from the IR camera to determine the 3D pose $(x, y, z, \phi, \theta, \psi)$ of the platform and a map of the target maintenance environment. The method furthermore extract significant thermal signatures from the environment and references these events with respect to the created map.

II. METHODOLOGY

The high-level functional block diagram of our approach is depicted in Figure 2. The proposed method for guiding, navigating and controlling the unmanned multi-copter consists of two main functions: the **tactical** Guidance Navigation and Control (GNC) function and the **strategic** GNC function [9].

The tactical GNC function makes optimal short-term trajectory planning decisions at a high update rate to accommodate a possible high speed of operation of the aerial vehicle in a potentially cluttered environment. These decisions are based on vehicle state (e.g. position, velocity, and attitude) updates from the navigation module and a local map of the environment. This map describes attributes of the local environment as perceived by the sensors that can be used in the path planning and collision avoidance process. Example maps in the context of this paper are forest maps and time-to-contact maps.

The strategic GNC keeps track of the planned and actual flight trajectories and provides the pilot with the next waypoint(s) that must be reached. In unknown environments, the strategic GNC function keeps track of a rough estimate of the platform trajectory and associated global map of the environment. Based on this information and the goals defined by the specific mission (i.e. outdoor/indoor navigation and mapping mission), this function will provide the tactical GNC algorithms with constraints regarding its path planning decisions (i.e. preferred direction of travel, decision between multiple similar choices such as doors or windows) and associated control behaviors. The global map can be formed using various typical SLAM back-end methods like composition of the global map from multiple local maps using for example Hierarchical SLAM [10] or GraphSLAM methods (e.g. [11]).

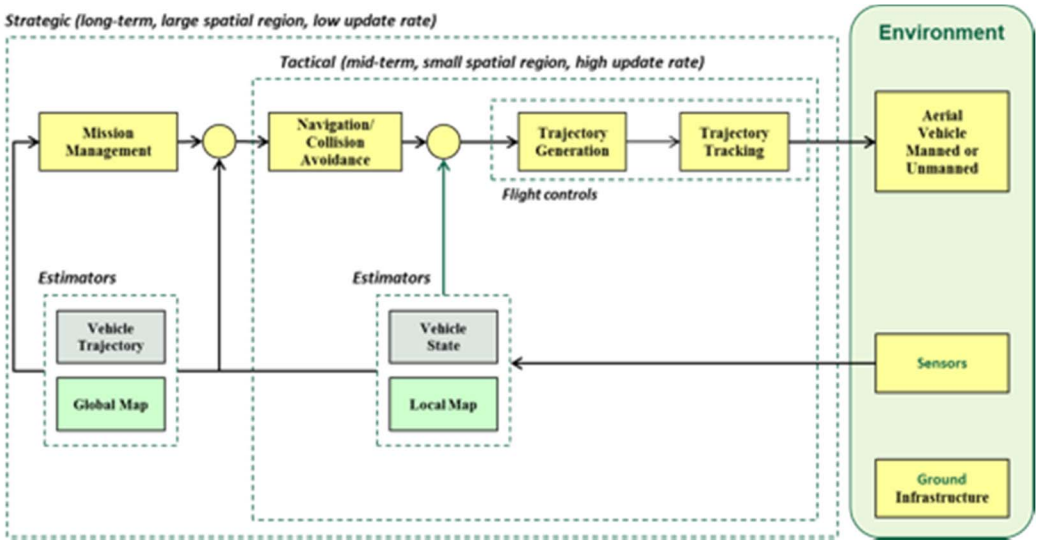


Figure 2. High-level functional block diagram of the hierarchical guidance, navigation and control function.

A block diagram of the proposed tactical GNC navigator and mapping method is shown in Figure 3. Blocks (a) and (b) take accelerometer and gyro measurements from the onboard IMU and perform the attitude and position calculations, solely based on the IMU measurements using the method described in [12]. The point cloud provided by the 3D imager is expressed in the body frame and will be first converted to the navigation frame using a best estimate of the attitude. Then, features are extracted from the point cloud in (c) and associated from frame to frame in (d). As the environment is structured, typical 3D structures will be used such as planar surfaces, pipes, corners or edges. The displacement and/or relative orientation change from frame to frame can be used to estimate the delta-position, $\Delta\tilde{\mathbf{r}}^n_{DEP}$, and delta-orientation, $\Delta\tilde{\mathbf{C}}^n_b$, of the platform. The same can be obtained from the IMU mechanization ($\Delta\tilde{\mathbf{r}}^n_{IMU}$,$\Delta\tilde{\mathbf{C}}^n_b$), and both can be input to a Complementary Kalman Filter (CKF) with error state vector:

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$$\mathbf{x}_1 = [\delta\mathbf{v}^n \quad \delta\boldsymbol{\psi}_{nb} \quad \delta\boldsymbol{\omega}_{ib}^b \quad \delta\mathbf{f}_b] \qquad (1)$$

where $\delta\mathbf{v}^n$is the velocity error in the NED frame, $\delta\boldsymbol{\psi}_{nb}$is the miss-orientation vector, $\delta\boldsymbol{\omega}_{ib}^b$is the gyro bias vector, $\delta\mathbf{f}_b$is the specific force bias error. After the CKF update step, the error states are fed back to the attitude and navigation computation blocks to reduce the drift errors. The measurement vector formed as follows:

$$z_{INS} = \Delta\tilde{\mathbf{r}}_{INS}^n - \Delta\tilde{\mathbf{r}}_{DEP}^n \qquad (2)$$

The error states in equation (1) only concern the IMU error states, and current implementation assumes that the errors from the 3D imager based translation motion estimation are Gaussian in nature. This is, however, not the case and will be addressed in future papers.

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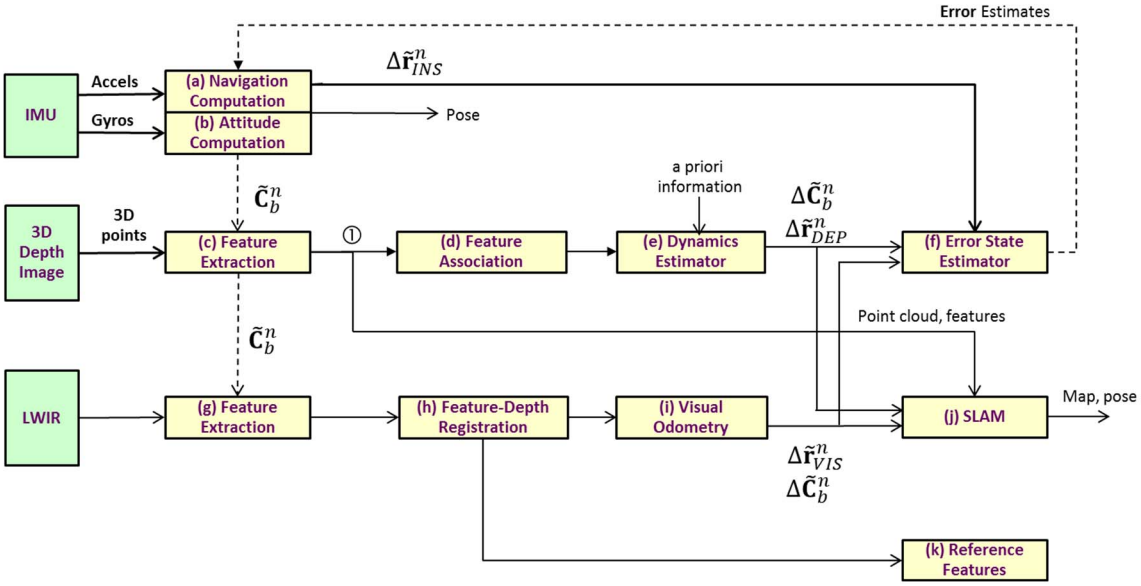


Figure 2. Functional block diagram of the proposed navigation and mapping method.

Although the LWIR camera is a low resolution sensor (80-by-60 pixels), features can still be extracted from the resulting imagery and is done so in block (g). Feature examples are corners, lines, or blobs. The lower resolution will result in a larger uncertainty of the feature location and, thus, larger estimation covariance when using these features for visual odometry [13]. Typical visual odometry results in an estimate of the change in orientation between the two image frames, and an estimate of the translational motion up to a scale factor due to unknown depth. The latter issue can be resolved by associating the LWIR sensor pixels with pixels of the 3D imaging sensor, and using the associated point cloud coordinate to obtain the unknown depth. The resulting translational motion can then input to the CKF of (f) by forming the following error measurement:

$$z_{LWIR} = \Delta\mathbf{r}_{INS}^n - \Delta\mathbf{r}_{LWIR}^n \qquad (3)$$

It is important to include good estimates of the error covariance matrices Σ_{LWIR} and Σ_{DEP} to reflect the difference in quality of both error measurements.

In addition to an error estimator to keep the inertial drift contained, the proposed method uses the odometry estimates from the LWIR camera and the 3D imager, the point cloud measurements and the extracted features and input these to a 3D SLAM algorithm. Since the objective is to generate a position solution and map in real-time, this SLAM methods must be fast and robust. The method here is based on the methods proposed in [15] and [16], where 3D features are extracted and subsequently used to perform odometry and generate a LIDAR map. The main difference between our method and the method proposed is the manner in which features are extracted and used to perform the odometry step. An alternate method is described in [5] and [14]. The SLAM method described in those papers consists of two stages; the first stage run open loop and uses the pairwise Iterative Closest Point (ICP) algorithm [15] to estimate the incremental pose. When loop closure is detected (earlier environment is encountered a second time), it uses an extension of Lu and Milios laser scan matching algorithm [18] to find the change in pose.

The final step in our methodology (k), takes the features extracted from the LWIR images and maps them on the 3D environment using a direct mapping from the LWIR image frame to the depth image frame and the map.

III. Hardware Setup

Throughout the remainder of the paper preliminary results will be shown based on data collected in the maintenance tunnel environment shown in Figure 1. The hardware platform and the location of the sensors with respect to the platform can be seen in Figure 3. The small-size UAS platform is a Blackout

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quadcopter equipped with a 3DR Pixhawk flight controller and an Odroid XU4 onboard computer running Ubuntu. The Robotics Operating System (ROS) is used to interface with and integrate all the sensors and control the flight controller.

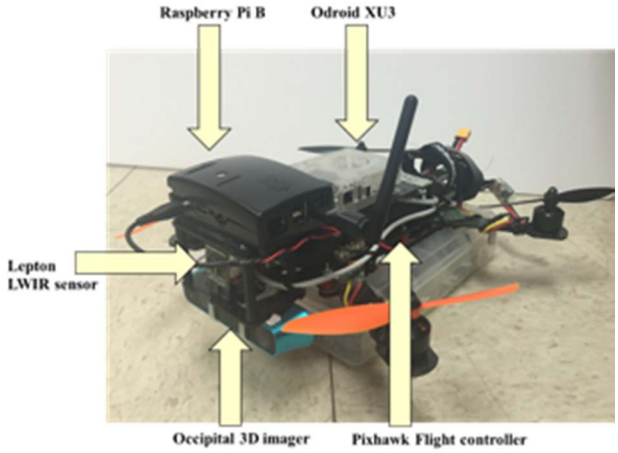


Figure 3. Quadcopter platform: sensor payload overview.

The optical sensors include (i) a forward-pointing Occipital providing both short-wave infrared (SWIR) imagery (not used in this paper) and depth information, (ii) and a FLIR Lepton LWIR camera to pick up heat signature image of objects in the environment. The latter is connected to a Raspberry Pi via a SPI interface; the Raspberry Pi send the LWIR frames to the Odroid where they are processed using a ROS software module (i.e. node). For diagnostic purposes and communication with each of the sensors during flight, an 802.11b wireless connection is established between the Odroid and a ground station. During data collections, all of the desired sensor data is recorded on the Odroid, stored in "rosbags", and then offloaded to the ground station where ROS is used to analyze the data via it play-back capability. The hardware block diagram of the system is shown in Figure 4.

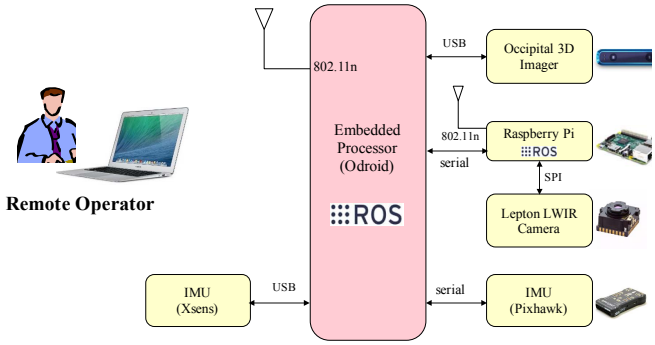


Figure 4. Hardware block diagram of test setup.

IV. Point Cloud-based Dynamics Estimation

An example of the point cloud output by the 3D imager in a relevant environment and down-sampled using a voxel grid with voxel size 4cm-by-4cm-by-4cm is shown in Figure 5. It is clear from this figure that the environment can be considered structured. Therefore, the first step is to extract relevant features from the point cloud such as planar surfaces and cylindrical objects. Various algorithm are available and have been discussed in, for example, references [19] and [20].

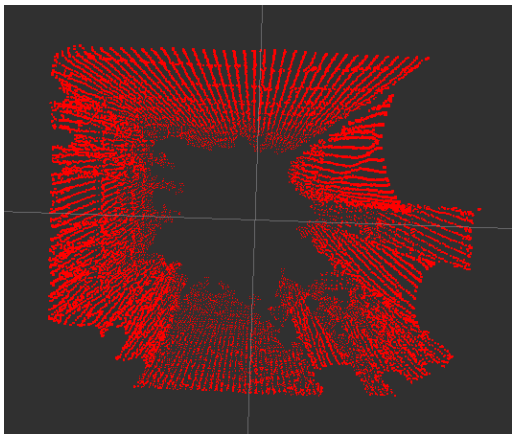


Figure 5. Example of point cloud at one time epoch.

After association of the planar surfaces from one frame to the next, the closest distance to each planar surface 'j', ρ_j, and the normal vector, $\mathbf{n}_j$, of each plane can be estimated from the point cloud points associated with each planar surfaces. The odometry value $\Delta\mathbf{r}$ is related to these parameters as follows:

$$\Delta\mathbf{r}\cdot\mathbf{n}_j(t_i)=\rho_j(t_{i+1})-\rho_j(t_i)=\Delta\rho_j \Rightarrow$$
$$\underbrace{\begin{bmatrix}\mathbf{n}_1^{\mathrm{T}}(t_i)\\ \vdots\\ \mathbf{n}_N^{\mathrm{T}}(t_i)\end{bmatrix}}_{\mathbf{H}}\Delta\mathbf{r}\cdot=\underbrace{\begin{bmatrix}\Delta\rho_1\\ \vdots\\ \Delta\rho_N\end{bmatrix}}_{\Delta\boldsymbol{\rho}}\Rightarrow \quad (4)$$

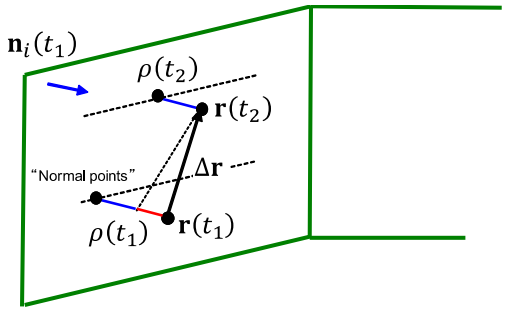


Figure 6. Planar surface-based navigation geometry.

If three non-parallel planar surfaces are available, then this system of equations can be solved to obtain the translational motion via an ordinary least squares solution:

$$\Delta\hat{\mathbf{r}}=(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\Delta\boldsymbol{\rho} \quad (5)$$

Given the planar surface normal vectors, estimates for the rotational motion can be obtained as well.

$$\mathbf{C}_{t_i}^{t_{i+1}}\mathbf{n}_j(t_i)=\mathbf{n}_j(t_{i+1})\Rightarrow$$
$$\hat{\mathbf{C}}_{t_i}^{t_{i+1}}=\arg\min\left\{\sum_{j=1}^{N}\mathbf{C}_{t_i}^{t_{i+1}}\mathbf{n}_j(t_i\cdot\mathbf{n}_j(t_{i+1})\right\} \quad (6)$$

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This estimate only needs two non-parallel surfaces and would work if one of the side walls and floor or ceiling is available. A method to solve equation (11) using quaternions is provided in [6].

When three non-parallel surfaces do not exist, the values for ρ_j and $\mathbf{n}_j$ and can still be used in the CKF mechanization of a tight laser-inertial integration approach by forming:

$$z_{NCP} = \Delta\boldsymbol{\rho}_{INS} - \Delta\boldsymbol{\rho}_{DEPTH} \qquad (7)$$

where $\Delta\boldsymbol{\rho}_{INS}$ is synthesized using equation (4) by substituting $\Delta\boldsymbol{r} = \Delta\mathbf{r}_{INS}$.

An alternate method to obtain estimates of the translational motion and rotation is the use of ICP [17]. Using two consecutive full point clouds would require a significant amount of computation time, however, using only the points identified to be on the planar surfaces or by using the reduced voxel grid the computation time could be significantly reduced.

Alternatively, line and point features in 3D can be identified and associated from one frame to the next based on the difference between their projected location (using the translation and rotation estimates of (a) and (b) and associated uncertainty gates) and their observed location in the next frame. The odometry (translation) can be directly observed form the point features:

$$\textbf{Points:} \quad \Delta\hat{\mathbf{r}} = \mathbf{p}_j(t_{i+1}) - \boldsymbol{C}_{t_i}^{t_{i+1}}\mathbf{p}_j(t_i) \qquad (8)$$

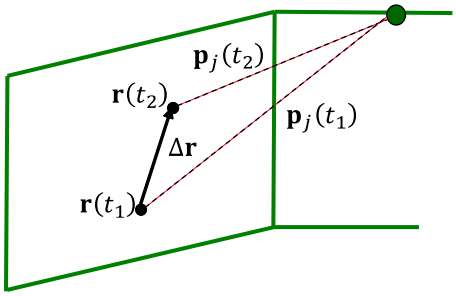


Figure 7. Point -based navigation geometry.

In case of the line features, their orientation can be used to estimate the rotational motion between two consecutive frames in a manner similar to equation (11). An estimate of translation cannot be directly observed from one line segment as there is an ambiguity in the direction of the line, $\boldsymbol{\ell}_j$, and in a cylinder around the line. However, if two or more non-parallel lines are observed and rotated to align, an estimate of the translation can be obtained by finding the translation that maps the points on the lines in the next frame on the current frame.

Figure 8 shows the simulation results for the proposed integration strategy using only planar features and inertial measurements.

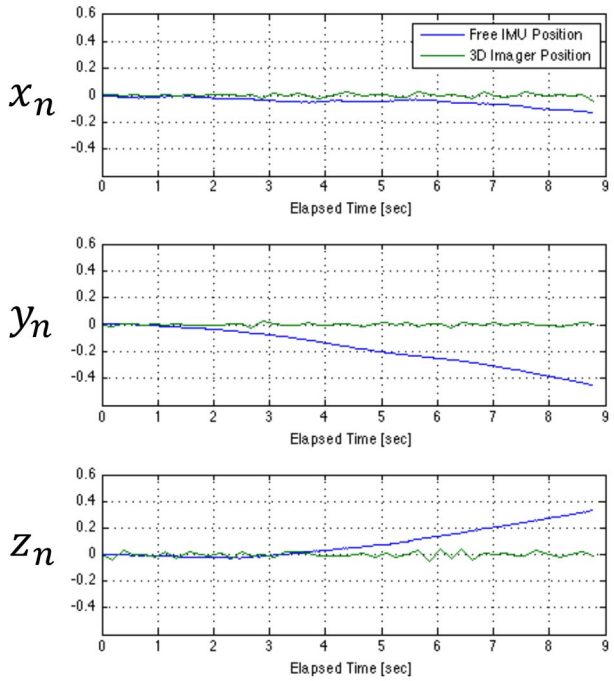


Figure 8. Simulation results of tightly-integrated 3D imager-inertial navigation.

V. Point Cloud Feature Extraction

Various methods and software tools exist to extract 3D features from a point cloud. For example, Figure 9 shows the planar surfaces extracted from one frame collected using the hardware platform and feature extraction methods from the PointCloud library [19]. The walls, floor and ceiling can be easily identified as well as the cylindrical pipe structures. The disadvantage of the method is its required computational resources.

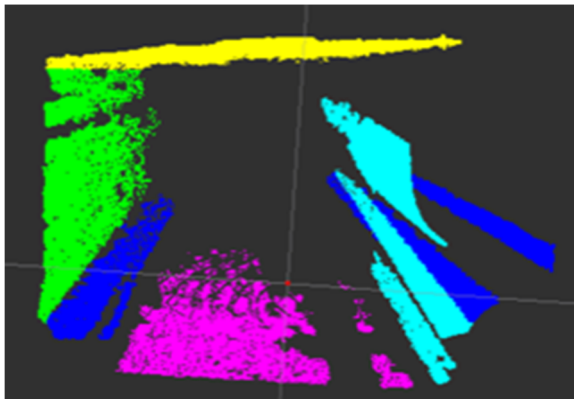


Figure 9. Feature extraction results.

An alternative method that reduces the number of points used for feature extraction is referred to as environment slicing and involves taking a set of slices at discrete depths (point cloud z-values), computationally low-cost operation. An example result of this slicing process is shown in Figure 10. Next, 2D

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features (e.g. lines, points, and arcs) can be extracted and combined across the multiple slices to from 3D features (e.g. planar surfaces, lines, and cylinders). For example, the ceiling surface will result in multiple line segments in the consecutive slices, that can be combined to form a planar surface. If necessary, a region growing operation can be performed to include the earlier omitted point cloud points.

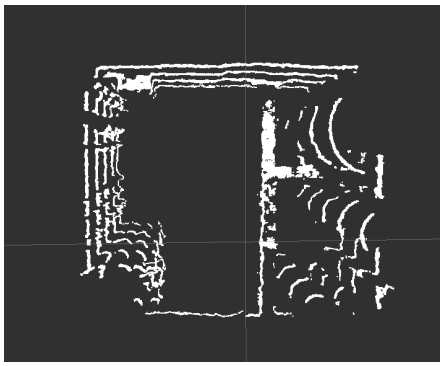


Figure 10. Environment slicing.

To increase the number of observables, an edge and point feature extraction method has been implemented using the depth image. The proposed method compares all points of the depth image (z-components of point cloud coordinates) to the points in their immediate 3-by-3 neighborhood. The point is considered an edge point if a depth change larger than a predefined value is detected between the point under consideration and any of its neighbors. From these edge features, line segments and point features can be identified using an 3D extension of component labelling. An example is of this process is shown in Figure 11.

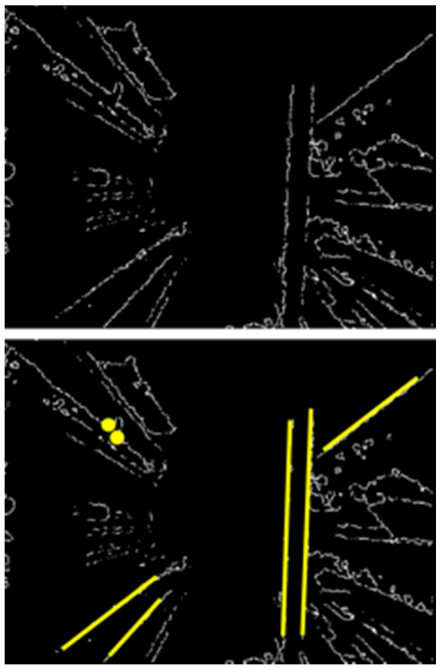


Figure 11. Edge and point feature extraction.

VI. LWIR Feature Extraction

The main function of the onboard LWIR sensor is to find significant variations in the thermal signature of the environment, that would indicate air/gas leaks, bad insulation, or other phenomena with a thermal signature. An example LWIR image frame is shown in Figure 12.

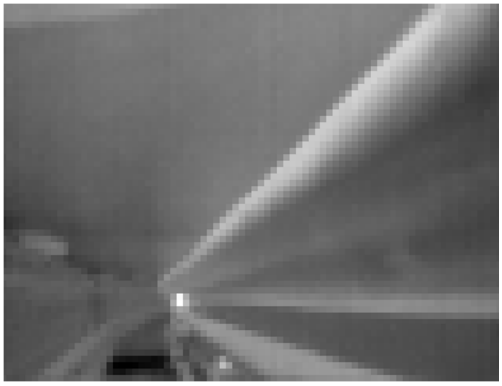


Figure 12. Example LWIR image frame with visible "hot" spots at the pipe joints.

The proposed method feature extraction method is shown in Figure 13. The edges of the pipes in the maintenance tunnel are obtained using a Laplacian of Gaussian (LoG) filter [21].

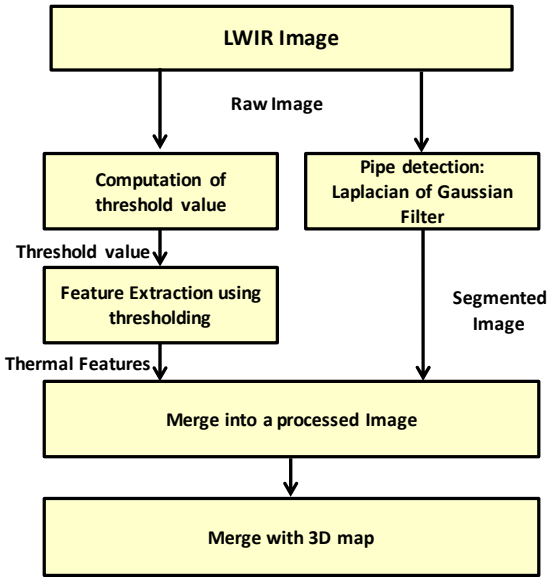


Figure 13. Functionl block diagram of LWIR feature extraction method.

An example of an extraction result is shown in Figure 14. The result is clear, but very course due to the limited resolution of the LWIR sensor.

Since the Lepton LWIR is a thermal imager, "hot" spots can be observed and isolated using a straight-forward thersholding operation. However, as the thermal signature of the environment changes while flying through the tunnel, it will be

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necessary to compute the theshold dynamically. The method to compute the threshold used here was proposed in [22] and normalizes the height and dynamic range of the intensity histogram.



Figure 14. Example of pipe feature extraction.

Figure 15 shows the combined result of the thresholding operation and the pipe edge extraction method. The "hotter" pipe and joint can be clearly observed in this image.



Figure 15. Example of thermal signature extraction.

The last step in the process references the extracted hot spots to the map generated by the SLAM method.

VII. Flight Environment Considerations

The platform described in Section 3 was "flown" manually through the maintenance tunnel. All previous figures were derived using data collected during this flight. During the flight it was observed that dust clouds were generated caused by the multi-copter's rotors' wake. This is illustrated in Figure 16. No significant decrease in the quality of the 3D imager point clouds or the LWIR images was observed.

VIII. Summary and Conclusions

This paper introduces a method for GNSS-denied navigation and mapping in a structured tunnel environment. A multi-copter platform equipped with the aforementioned sensors has been designed, built and flight-tested in an appropriate environment. Preliminary analysis of the collected data shows successful extraction of the features that will be used in the integrated navigation algorithm. To get insight in the absolute performance of the algorithms, a truth reference environment will be setup by mapping the tunnel environment with a terrestrial laser scanner.

Current and future work include the evaluation of the navigation algorithms using the collected data, the development of an autonomous real-time planning, flight and mapping capability, and the development of collision avoidance strategies.

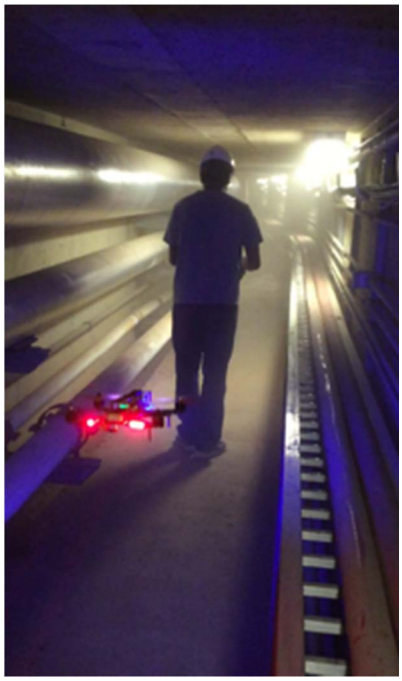


Figure 16. Dust generated by the multi-copter in flight.

Acknowledgments

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